Introduction

Prerequisites

- Room Access
    - Safety Instruction
- GPU-Cluster Access
- CPU-Server / GPU Server
- Vivado Server
- Thin Clients Access

Introduction

- Thesis Basics
    - Latex Thesis Template
    - Allowed Tools
    - Technical support during your thesis
- Scientific Work
- Scientific Writing
- AI Resources
- I upload everything to here

Thesis Basics

- Allowed Tools
- Latex Thesis Template

Latex Thesis Template

Use only the template

Allowed Tools

You can use AI, but you need to mark it. Lets have a look at the guideline

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Technical Support during your thesis

I will only help on mac os or linux. If you use windows and run into problems, it is your problem.

Scientific Work

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1. Clear Research Question

- Described by the supervisor
- A good scientific work starts with a clear and well-defined research question. This question should be specific, relevant, and testable.

Bad Example

"Can the use of quantization enhance the neural networks?"
What is missing?

- Quantization scheme
- Quantization for what?
- how does it get better?
- What is the target platform?

Good Example

"Can the use of fixed-point quantization of a neural network for all parameters and calculations reduce the computational overhead (memory, speed) on MCUs while achieving comparable performance during the training?"

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2. Thorough Literature Review

A thorough literature review is essential to understand the current state of knowledge on the research topic. It helps to identify gaps in existing research and provides a foundation for the new study.

Bad Example

- Looking into one source only that has not going through a peer-review process
  - Wikipedia
  - Arxiv.org
- Don't look for disadvantages of technologies
- Save only papers supporting your hypothesis

Ideal Example

- Have multiple peer-reviewed (ideally double blinded) sources for each statement itself
  - Find the root of a statement, i.e. if paper A cites something from paper B, use paper B as your source instead
- Also, includes papers that are against you hypothesis. Why are they against it? Or is it only for subparts of your hypothesis?

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3. Hypothesis

- From the research question, one can develop multiple hypothesis
- In an experiment we test the hypothesis not the research question
- Thus, the hypothesis should be falsifiable

"An educated guess that proposes a testable explanation for an observed phenomenon."
• "Educated" (based on research),
•"Guess" (proposed explanation),
•"Testable" (can be proven true/false) around it.

(Default): The use of 16bit linear quantisation instead of floating point will not reduce the memory consumption.
/ : The use of 16bit fixed-point quantization instead of floating point will reduce the memory consumption by 50%.

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4. Well-Designed Experiment

A well-designed experiment is crucial to collect reliable and valid data. This includes selecting appropriate methods, instruments, and sampling techniques.

Bad Example

- Having a script and changing some code parts and some numbers.
- When you get a good result, write down the experiment configuration.
- Forget some variables that influence the experiment heavily.

Good Example

- Write down the hypothesis you want to proof, what to test, how to test, which variables should be changed and why. Write down what you expect.
- Have a script to start the experiment with an experiment configuration as an argument.
- Be able to rerun all your experiments quickly, in case you find a bug shortly before the deadline.
- Save the experiment results and the configuration automatically in the folder.

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5. Transparent and Reproducible Methods

The methods used in the study should be transparent and reproducible. This includes providing detailed descriptions of the procedures and data analysis.

Bad Example

- Not being able to rerun the experiment.
- Show the good results, throw away bad results

Good Example 1

- Write down all experiments.
- Bad results are part of a good experiment.
- Save the git-hash with the experiment configuration.
- Set all random seeds by hand. If you use a framework look up, what they say about determinismn.
- Add a readme to the experiment folder of your git-repo that states, how you have run the commands.

Good Example 2

- State on which hardware you have run the experiment, as sometimes determinismn is only possible on the similar hardware.
- If you can not make the process behave deterministically, save everything you get back from the process.

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6. Objective and Impartial Analysis

The analysis of data should be objective and impartial. It should be free from bias and based on the data collected.

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7. Honest and Accurate Reporting

The results of the study should be honestly and accurately reported. This includes presenting both positive and negative findings.

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8. Respect for Ethics and Integrity

Good scientific work should be conducted with respect for ethics and integrity. This includes obtaining informed consent, avoiding harm to participants, and maintaining confidentiality.

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9. Continuous Improvement

Good scientific work is a continuous process of improvement. It involves revising and refining the research question, methods, and analysis based on new information and feedback.

Scientific Writing

In my experience it is mostly about breaking the writing blockade. I recommend doing something like the Zettelkastenprinzip.

Zettelkastenprinzip / Note Box Principle

1. Differentiate between the following steps clearly. Don't do them at the same time!
2. Read and understand.
3. Write down your thoughts about the text you read on a note (digital).
4. Categorize the note in the box, cross reference to other notes.
5. Prepare the text as a string of notes. Sometimes parts of the notes will be used multiple times within the text.
6. Reformulate the string of notes to paragraphs, then to chapters. Revise the thesis over and over.

Iterative process 1: Writing

1. Write down thoughts as bullet points / short sentences.
2. Arange a set of sentences to a paragraph.
3. Arange a set of paragraphs to a chapter.
4. Arange a set of chapters to a thesis.

Most Important: Only do one step at a time!!!

- Make this an calendar appointment for 20mins for one points.

Iterative process 2: Reformulating

1. Reformulate a bullet point
2. Reformulate a sentence
3. Reformulate a paragraph
4. Reformulate a chapter.

Most Important: Only do one step at a time!!!

- Make this an calendar appointment for 20mins for one point.

AI-Sources

- Theory
  - Deep Learning (Ian Goodfellow, Yoshua Bengio and Aaaron Courville)
  - GKI Lecture held by me (but I will probably also give a short introduction to neural networks at some point.)
  - Stanford University Lecture (Sort them by the most popular ones.)
- Programming
  - Pytorch Tutorials

Tools I use

- Obsidian for Zettelkasten and Note Taking
- Jabref for Bibtex (but can not recommend it, Zotero & Citavi are likely the better option)
- CLion (but might swap to VSCode at some point in time)
- JJ (the better git)
- Overleaf

Next Time

- Read "the hypothesis Fall from Nelson Baloian" as it is a good additional resource
- Prepare the research question as good as you can. If possible find important literature and formulate a hypothesis (it is not a problem, if it feels wrong. The early you bring something up, the earlier we can have a look at it)
- Testing
    - Unit Integration System
    - TDD
- CleanCode Cheat Sheet

Next Next Time

- The ElasticAi.Runtime.EnV5 AI Framework
- The ElasticAi.Creator.quantizedGrads
- AI Training?

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https://nc.buron.science/index.php/s/jBFdPpkt3rxJNsW